

Fares Mohammed Nabil Elsherif

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PROFESSIONAL SUMMARY

Communications & Information Engineering student at Zewail City of Science and Technology, with proven hands-on experience designing, building, and debugging hardware/software integrated control systems using Arduino, ESP32, and C++. Demonstrated ability to engineer real-time closed-loop systems, industrial circuits, and sensor-actuator pipelines that deliver measurable results — from eliminating timing drift to cutting manual intervention to zero. Deeply motivated by industrial automation and mechatronics, actively seeking an internship to contribute immediately and develop professional-grade engineering skills.

TECHNICAL SKILLS

Languages:	C++ (OOP, STL), Python (GUI, scripting)
Machine Control:	Machine control software, closed-loop control, real-time systems
Industrial Automation:	Sensor-actuator integration, automated control, process automation
Circuit Design:	Industrial circuit design, NE555, CD4017, I2C, amplifiers, voltage regulation
Embedded Systems:	Arduino, ESP32, hardware/software integration, edge-based signal detection
Tools & Languages:	Git, GitHub, signal processing, documentation Arabic (native), English (professional)

PROJECTS

Smart Traffic Light Control System | *ESP32 · NE555 · CD4017 · I2C · C++*
github.com/s-fares-nabil/Smart-Traffic-Light-System-using-555-Timer-CD4017-ESP32-and-LCD

- Designed machine control software and hardware for a 3-phase traffic light system using NE555 timer IC and CD4017 decade counter, eliminating software timing drift through industrial circuit design.
- Programmed ESP32 for real-time signal monitoring across 3 channels with edge-based detection; reduced state-mismatch errors to zero across all test cases.

Multi-Sensor Fish Farm Monitoring & Control System | *Arduino · C++ · Real-Time Control*

- Engineered machine control software for a real-time closed-loop system integrating 4 sensor types (temperature, pH, dissolved oxygen, flow); applied signal processing to improve accuracy by ~30%.
- Designed system integration architecture for continuous unattended operation, reducing manual intervention from hourly to zero during normal operation.

Smart Irrigation & Water Management Control System | *Arduino · Python · Sensor-Actuator Integration*

- Built machine control software governing 3 operational states; threshold-triggered pump actuation within 2 seconds of breach, validated across 15+ scenarios — cutting estimated water usage by ~40%.
- Integrated Python mobile interface for remote monitoring and manual override, extending usability beyond physical hardware.

Electronic Circuit Design Portfolio | *Industrial Circuit Design · Debug · Documentation*

- Designed, built, and debugged 5 functional circuits (DC power supply, traffic light controller, diode logic gates, amplifier); resolved 3+ design faults through fault isolation; produced full annotated documentation portfolio.

EDUCATION

Zewail City of Science and Technology — B.Sc. Communications & Information Engineering	2025 – 2029
Coursework: Calculus, Physics, OOP, Electronic Circuits, Engineering Design	
El-Sadat STEM High School — STEM Track 89%	Graduated 2025

UNIVERSITY ACTIVITIES & CERTIFICATES

Usher, Zewail City Career Fair 2026
Guided attendees and coordinated logistics

IEEE Student Branch
Embedded systems workshops & collaborative problem-solving

Volunteer, Community Development Association
Event organization and coordination

Digital Egypt Cubs Initiative (DECI)
Ministry of Communications (MCIT), 2024

Basics of Cybersecurity
National Telecommunications Institute (NTI), 2024